

Recursive prediction for long term time series forecasting using advanced models

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Abstract

There exists a wide range of paradigms, and a high number of different methodologies that are applied to the problem of time series prediction. Most of them are presented as a modified function approximation problem using input/output data, in which the input data are expanded using values of the series at previous steps. Thus, the model obtained normally predicts the value of the series at a time $(t + h)$ using previous time steps $(t - \tau_1), (t - \tau_2), \dots, (t - \tau_n)$. Nevertheless, learning a model for long term time series prediction might be seen as a more complicated task, since it might use its own outputs as inputs for long term prediction (recursive prediction). This paper presents the utility of two different methodologies, the TaSe fuzzy TSK model and the least-squares SVMs, to solve the problem of long term time series prediction using recursive prediction. This work also introduces some techniques that upgrade the performance of those advanced one-step-ahead models (and in general of any one-step-ahead model), where they are used recursively for long term time series prediction.

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1. Introduction

There are several methodologies to deal with the problem of time series prediction [23]. Neural networks, fuzzy systems, support vector machines, AR-ARMA models, etc., are among the paradigms that have been applied to that problem. In the case of long term time series, two trends can be taken to tackle the prediction problem: direct prediction and recursive prediction [10]. Direct prediction implies the construction of different models for the different prediction horizons needed. On the other side, recursive prediction only uses one model to predict all the horizons needed, but with the drawback that the error committed in nearest horizons can be transmitted to further horizons. This paper deals with recursive prediction, showing two different methodologies to build the recursive model, and presenting a general technique that improves the prediction accuracy for this type of problem.

The methodologies used in this paper are the TaSe Takagi–Sugeno–Kang (TSK) fuzzy model [5,9,8] and the least-squares-support vector machines (LS-SVMs) [21]. The first is an advanced fuzzy learning methodology that provides an understandable and interpretable model, given a set of input/output (I/O) data. It obtains a good performance in function approximation and time series prediction problems, comparing to other soft-computing techniques. The second is a reformulation of the well-known SVM model, which is based in the statistical learning theory. It provides high performance with good generalization in classification and function approximation problems. This paper does not intend to compare these different methodologies, but rather it deals with the general problem of long term time series prediction using recursive models (independently on the paradigm used), and with a general technique to obtain a better performance.

The TaSe model, which was first presented in [5] and whose full learning methodology was fully developed in [9], is a modified Takagi Sugeno Kang (TSK) fuzzy model [22] that uses a grid partitioning (thus a grid-based fuzzy system (GBFS) [17] type) of the input space. The TaSe model

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partially overcomes the two main drawbacks of TSK fuzzy models: the curse of dimensionality problem and the lack of interpretability. The curse of dimensionality problem has to do with the exponential growth in the number of rules in this type of fuzzy models, which occurs when the number of input dimensions and the number of membership functions (MFs) per input variable grow up. The lack of interpretability has to do with different aspects that TSK fuzzy models should include, but those are generally lost in the designing and in the learning processes: the transparency of the model (transparent partition of the input space), the need of maintaining a relatively low number of rules [4] (very related thus with the curse of dimensionality problem) and, more specifically for the TSK models, the interaction of the global and local models [11] (that should allow us to interpret the rules comprising the TSK system as linearizations of the nonlinear subjacent system). The TaSe model considers these intrinsic aspects of the TSK fuzzy systems, providing high accuracy for function approximation and time series prediction problems [8,9]. It also provides full interpretability to the rule consequents, using a fundamental key in the field of function approximation theory and mathematical analysis, as is the Taylor Theorem. The TaSe model allows us to interpret its polynomial rule consequents as the Taylor series expansion of the model output around the rule centers, thus giving an explicit interpretation of the subjacent model.

A different type of model is the LS-SVMs. LS-SVMs are reformulations to standard SVMs that lead to solving linear Karush–Kuhn–Tucker (KKT) systems [20]. LS-SVMs are regularized supervised approximators, closely related to regularization networks and Gaussian processes, but that additionally emphasize and exploit primal–dual interpretations from optimization theory [21]. LS-SVMs, as well as SVMs, suffer also from the curse of dimensionality, but in the number of training data points. Fixed-size LS-SVM [21] is one of the approaches that have appeared in the literature to tackle this problem, which is one of the main areas of research for this paradigm.

The rest of the paper is organized as follows: Section 2 introduces the main concepts related to the TaSe methodology and, as an example, it presents its application to the well-known Mackey–Glass time series benchmark [15]. Section 3 reviews the LS-SVMs methodology. It also includes as an example its application to the Mackey–Glass time series. Note that since both methodologies are very different learning methodologies, the application example has been included as a subsection of each respective section. However, each subsection also includes a comparison with other methodologies, highlighting the main advantages of the respective model. Section 4 briefly presents a modified point of view of the time series forecasting problem for long term prediction problems. Section 5 presents a general modified learning methodology, showing its application for the TaSe and LS-SVM models, to improve the prediction of long term time series. Finally, Section 6 concludes the paper.

2. The TaSe model learning methodology

In general, the structure of a multiple-input single-output (MISO) TSK [22] fuzzy system and its associated fuzzy inference method comprise a set of K IF–THEN rules in the form

$$\begin{aligned} \text{Rule}^k : & \text{IF } x_1 \text{ is } \mu_1^k \text{ AND } \dots \text{ AND} \\ & x_n \text{ is } \mu_n^k \text{ THEN } y = R^k, \end{aligned} \quad (1)$$

where the μ_i^k are fuzzy sets characterized by MFs $\mu_i^k(x_i)$ in universes of discourse U_i (in which variables x_i take their values), and R_k are the consequents of the rules.

The output of a fuzzy system with rules in the form shown in Eq. (1) can be expressed (using weighted average aggregation) as

$$F(\vec{x}) = \frac{\sum_{k=1}^K \mu^k(\vec{x}) R^k}{\sum_{k=1}^K \mu^k(\vec{x})}, \quad (2)$$

provided that $\mu^k(\vec{x})$ is the activation value for the antecedent of the rule k , which can be expressed as

$$\mu^k(\vec{x}) = \mu_1^k(x_1) \mu_2^k(x_2) \dots \mu_n^k(x_n). \quad (3)$$

Now the peculiarities and specifications of the TaSe model, as well as its learning algorithm, are briefly reviewed. For a complete presentation refer to [9].

2.1. TaSe specifications

The TaSe model uses rule consequents (R^k in (1)) in the form

$$\begin{aligned} f(\vec{x}) = & f(\vec{a}) + (\vec{x} - \vec{a})^T \left[\frac{\partial f}{\partial \vec{x}_i}(\vec{a}) \right]_{i=1, \dots, n} \\ & + \frac{1}{2} (\vec{x} - \vec{a})^T W (\vec{x} - \vec{a}) + \dots \end{aligned} \quad (4)$$

(where $\vec{a}$ is the rule center specified in the respective rule antecedent), i.e., in the form specified by the Taylor Theorem to approximate any function $f(\vec{x})$, around a point $\vec{a}$, as a summation of polynomials.

Nevertheless, the fact of using rule consequents in this form does not guarantee that the rules are interpretable using the Taylor Theorem. In general, the consequent of a rule not necessarily has a direct relationship with the output of the model in the area covered by the rule [9]. The TaSe model allows to interpret the rule consequents, as the Taylor series expansion of the model output with respect to the rule centers, thanks to two main properties. First, the model output is continuous and differentiable at least as many times as the order of the polynomial form used in the consequents. This way, the Taylor Theorem can be applied to the output function of the model (Eq. (2)). And second, the activation value of every MF is zero at each rule center, except for the MFs of the respective rule. This way, the respective rule consequent is the only polynomial function activated for the respective rule center. These requirements

are fulfilled thanks to the use of special type of MF bases, called orderly local membership function bases (OLMF bases), which were first presented in [1]. The main characteristics of the OLMF bases are:

- All MFs are local (i.e., non-negative and vanishing with the distance), defined in a delimited domain and of the same type.
- Every MF extreme point must coincide with the center of the adjacent MF (they form a partition, thus avoiding uncontrolled overlapping of the MFs).
- All MFs are p times differentiable and the p th derivative of the MF is continuous in all its domain.
- The p th derivative of the MF vanishes at its center and at its boundaries.
- The basis must accomplish the addition-to-unity property.

To construct an OLMF type, we simply use Hermite interpolation with the previous requirements specified as equality constraints for the 0th... p th derivatives, thus getting a spline-type MF basis [9]. p is also the order of the polynomial rule consequent (as in Eq. (4)). The TaSe model uses these OLMF bases for the input space grid partitioning and rule consequents resembling the Taylor series formulation in Eq. (4). Thanks to these characteristics, *the rule consequent of each rule k can be interpreted as a truncated Taylor series expansion of order p of the output of the fuzzy system $F(\vec{x})$ about the point $\vec{x} = \vec{a}_k$, the center of the k th rule [1,9].* This way, the TaSe model thus allows us to learn interpretable models for function approximation from a set of I/O data [9].

2.2. TaSe learning methodology

TSK fuzzy learning implies two parallel processes: structure identification (that deals with finding the optimal input space subdivision/number-of-rules according to the specified training I/O data) and parameter adjustment, both for rule antecedents and for rule consequents.

With respect to the parameter adjustment, the rule consequent coefficients can be linearly obtained using least-squares to solve the error function

$$J = \sum_{m \in D} (f(\vec{x}^m) - z^m)^2. \quad (5)$$

The rule antecedent parameters (specifically only the centers positions have to be optimized) will be optimized using the local-search Levenberg–Marquardt algorithm.

The structure identification approach used for the TaSe model uses an incremental iterative method. It will add one MF per iteration in the variable selected to be the optimal one to receive such MF. The structure identification algorithm stops whenever the validation error stops decreasing [9].

2.3. Application of the TaSe model to the Mackey–Glass time series prediction problem

The Mackey–Glass time series [15] is a very well-known time series benchmark. It is an artificial time series without noise, that has been widely used in the literature for neuro-fuzzy and other nonlinear model comparisons. This time series is described by the following delay differential equation:

$$\frac{dx(t)}{dt} = \frac{ax(t - \tau)}{1 + x^{10}(t - \tau)} - bx(t). \quad (6)$$

A sequence of 1200 data points were generated with an initial condition $x(0) = 1.2$ and $\tau = 17$, using the 4th order Runge–Kutta method. To make the comparisons with earlier works fair, the parameters were chosen so that the training vectors for the model have the following format:

$$[x(t - 18), x(t - 12), x(t - 6), x(t); x(t + 6)]. \quad (7)$$

Thus, after the training, a predictive model f would be obtained such that

$$\hat{x}(t + 6) = f(x(t - 18), x(t - 12), x(t - 6), x(t)). \quad (8)$$

For this application example, as well as for those presented in Section 5, we will use second order TaSe rules, i.e., the rule consequents will be second order polynomials in the form shown in Eq. (4), and $p = 2$ according to Section 2.1. The number of parameters per rule consequent in this case will be 15 (as can be obtained from Eq. (4), $1 + n + 1/2 * (n + n^2)$ with $n = 4$, where n is the number of input variables). From the 1200 data points generated, the last 1000 were taken, using 500 data for training and validation (randomly choosing 400 and 100, respectively, of the 500), and the final 500 for testing. Table 1 shows the evolution of the structure identification learning algorithm for an example execution (that iterates increasing the system complexity until the validation error stops decreasing). The algorithm starts with an initial MF configuration of 1 MF per variable, and iterates adding 1 MF to the selected variable. The execution time was 25 min in a P-800 MHz. However, it must be noted that for the incremental approach explained in Section 2.2, the computational cost highly increases as new MFs are added, and a threshold in

Table 1
Trace of the structure identification algorithm for the Mackey–Glass time series prediction problem

MFs					N rules RMSE (for predicting $x(t + 6)$)		
$x(t - 18)$	$x(t - 12)$	$x(t - 6)$	$x(t)$	–	Training	Validation	Test
1	1	1	1	1	0.032	0.029	0.034
1	2	1	1	2	0.0084	0.008	0.0098
2	2	1	1	4	0.0032	0.0037	0.0035
3	2	1	1	6	0.0024	0.0027	0.0028
3	2	1	2	12	0.0020	0.0024	0.0024
3	3	1	2	18	0.0014	0.0020	0.0017
3	3	1	3	27	0.0011	0.0018	0.0013

the maximum number of rules or parameters admitted should be set up in complex problems.

The learning procedure led to a system with 27 rules, each one with 4 antecedents (although note that there is only 1 MF in the input variable $x(t-6)$). The rule consequents are second order polynomials of 4 variables. The format of the 27 rules in this case have the form (being $(\vec{x} = \{x_1, x_2, x_3, x_4\} = \{x(t-18), x(t-12), x(t-6), x(t)\})$ and $y = \hat{x}(t+6)$):

Rule^k : IF x_1 is c_1^k AND x_2 is c_2^k AND x_3 is c_3^k AND x_4 is c_4^k THEN

$$y = a + (x_1 - c_1^k) * b_1 + (x_2 - c_2^k) * b_2 + (x_3 - c_3^k) * b_3 + (x_4 - c_4^k) * b_4 + \frac{1}{2}(\vec{x} - \vec{c}_k)^T C(\vec{x} - \vec{c}_k), \quad (9)$$

where $\vec{c}_k$ is the center of the k th rule. According to the characteristics of the TaSe model, this rule consequent is the Taylor series expansion of the model output (and thus of the expected approximated function) in the 4th dimensional space around the rule center $\vec{c}_k$. This implies that a is the value of the function in the point $\vec{c}_k, f(\vec{c}_k)$; the b_i are the partial derivatives in each of the 4 dimensions with respect to the rule center $\vec{c}_k, \partial f / \partial x_i(\vec{c}_k)$; and C is a matrix with the second order partial derivatives for the 4 dimensions with respect to the rule center $\vec{c}_k$. The number of rules needed to perform the approximation is low, since each rule comprises more information in its consequent. Each rule perfectly comprises the behavior of the function at the rule center.

Table 2 compares the prediction error for different computational paradigms presented in the bibliography for this benchmark problem (taken from [18]). In order to perform an equal comparison (also in complexity), the TaSe test error is shown for three MF configurations: $3 \times 2 \times 1 \times 1$ with 91 ($15 \times 6 + 1$ mobile center) parameters, $3 \times 2 \times 1 \times 2$ with 181 ($15 \times 12 + 1$ mobile center) parameters, and $3 \times 3 \times 1 \times 3$ with 408 ($15 \times 27 + 3$ mobile centers) parameters. The results drawn from this example show the convenience of the TaSe model, due to its high degree of accuracy for function approximation and time series prediction problems; with only 6 rules, the TaSe model is able to identify the nonlinear behavior of the Mackey–Glass time series, better than many other time series prediction methods proposed in the literature (with a similar model complexity). Furthermore, the consequents of these rules are the Taylor series expansion of the output of the model centered in their respective rule centers.

It must be noted now that using grid-partitioning of the input space for time series prediction has the disadvantage that since normally the data points are distributed around certain areas of the input space, there is a certain loss of efforts by equally covering the whole input space. This has two important consequences. First, there is an extra effort

Table 2

Comparison results of the prediction error of different methods for prediction step equals to 6 (500 training data)

Method	Prediction error (RMSE)
Auto regressive model	0.19
Cascade correlation NN	0.06
Back-prop. NN	0.02
6th-Order polynomial	0.04
Linear predictive method	0.55
Kim and Kim (Genetic algorithm and fuzzy system)	5 MFs 0.049 7 MFs 0.042 9 MFs 0.038
ANFIS and fuzzy system (16 rules $\approx$ 100 par.)	0.007
Classical RBF (with 23 neurons $\approx$ 200 par.)	0.0114
PG-RBF [18] (with 12 neurons $\approx$ 200 par.)	0.0029
TaSe fuzzy system with 6 rules	0.0028
TaSe fuzzy system with 12 rules	0.0024
Optimal TaSe fuzzy system with 27 rules	0.0013
LS-SVM	0.0005

See [18].

to perform the approximation, which can lead to an unneeded increase in system complexity and to a decrease in performance. And second, with respect to the interpretability, some rules could be centered in points in the n -dimensional space that does represent real data points of the function. To solve these problems, a clustering-based partitioning would place the rules in the areas of the input space in which they are needed, thus solving these drawbacks. In [6,7], some preliminary results about a clustering-based fuzzy model that keeps the basics of interpretability and performance of the TaSe model are presented.

3. Least-squares-support vector machines

The LS-SVM model [21] is defined in its primal weight space by

$$\hat{y} = \vec{\omega}^T \phi(\vec{x}) + b, \quad (10)$$

where $\vec{\omega}^T$ and b are the parameters of the model, $\phi(\vec{x})$ is a function that maps the input space into a higher dimensional feature space, and $\vec{x}$ is the n -dimensional vector of inputs x_j . In LS-SVMs for function approximation, the following optimization problem is formulated:

$$\min_{\vec{\omega}, b, e} J(\omega, e) = \frac{1}{2} \vec{\omega}^T \vec{\omega} + \gamma \frac{1}{2} \sum_{i=1}^N e_i^2, \quad (11)$$

subject to the equality constraints (inequality constraints in the case of SVMs)

$$e_i = y_i - \hat{y}(\vec{x}_i), i = 1, \dots, N. \quad (12)$$

Solving this optimization problem in dual space, leads to finding the λ_i and b coefficients in the following solution:

$$\hat{y} = \sum_{i=1}^N \lambda_i K(\vec{x}, \vec{x}_i) + b, \quad (13)$$

where the function $K(\vec{x}, \vec{x}_i)$ is the kernel function defined as the dot product between the $\phi(\vec{x})$ and $\phi(\vec{x}_i)$ mappings.

In case we consider Gaussian kernels, the kernel function $K(\vec{x}, \vec{x}_i)$ takes the form

$$K(\vec{x}, \vec{x}_i) = \exp \left[- \left(\frac{\|\vec{x} - \vec{x}_i\|}{\sqrt{2}\sigma_i} \right)^2 \right], \quad (14)$$

where σ_i is the width of the kernel, which together with the regularization parameter γ is the hyper-parameters of the problem. Note that in the case in which Gaussian kernels are used, the models obtained resemble radial basis function networks (RBFN); with the particularities that there is an RBF node per data point, and that overfitting is controlled by a regularization parameter instead of by reducing the number of kernels [19]. In LS-SVM, the hyper-parameters of the model can be optimized by cross-validation. Nevertheless, in order to speed-up the optimization, a more efficient methodology by Lendasse et al. can be found in [13].

3.1. Application of the LS-SVM model to the Mackey–Glass time series prediction problem

Now the application of the LS-SVM model to the Mackey–Glass time series is presented. The conditions of the problem have been already exposed in Section 2.3. The first 500 data points have been used for training and the final 500 for testing. We have made use of the LS-SVM implementation in Matlab, LS-SVMLab [14], to evaluate the performance of the application of the LS-SVM model to the time series prediction problems present in this paper.

After cross-validation, the optimal hyper-parameters obtained are $\sigma = 0.655$ and $\gamma = 4864$. The training root mean square error (RMSE) obtained for the first 500 training data points is 0.00041, and the test RMSE obtained for the last 500 test data points is 0.00052. As can be seen, the performance of the LS-SVM model exceeds that of the models presented in Table 2. Remember that in LS-SVM there are only two hyper-parameters that have to be optimized by a local search procedure, the regularization parameter and the kernel width. The structure of the obtained LS-SVM model, although said to be non-parametric, resembles an RBF network with one neuron per training data point. Also, as mentioned before, the control of overfitting is performed by the use of the regularization parameter γ . These conditions provide a high flexibility in the model behavior to adapt to each particular problem, and as seen for this specific example, the precision in the prediction is excellent.

4. Long term predictions

As it was mentioned, there are two main trends when dealing with long term time series prediction: direct prediction and recursive prediction [10]. Due to the characteristics of each approach, it can be expected that direct prediction provides a better performance in prediction accuracy, since it uses one specific model for each desired horizon [10]. Nevertheless, it has two main drawbacks that could render the use of recursive approaches more desirable: on the one hand, the modelling and the time series behavior understandability vanish as several different models are needed, one for each prediction horizon. On the other hand, for very long prediction horizons as in [3,12], which needs a very high number of predictions, a direct prediction approach seems useless since several different models would be required. Still for wider horizons, when the noise of the series is large, a direct prediction approach would be desirable to obtain better predictions. But lower levels of noise, a recursive prediction might be able to obtain good predictions using a single, and thus globally, more understandable model.

4.1. Recursive prediction

Now the problem of recursive prediction is studied more deeply. So far, two different approaches have been presented for modelling a set of I/O data in which the objective (for example for the Mackey–Glass time series) is to predict the output $\hat{x}(t+6)$, assuming the inputs specified in Eq. (7) are known. But for long term prediction problems, the objective could be expressed as: “having a short sequence of data, try to predict the rest of the data sequence until a certain time-step is reached”. Thus the training data for this model would be that of Eq. (7), but in fact, the application of the model to predict the complete data sequence should use iteratively the training data

$$\hat{x}(t+6) = f(\hat{x}(t-18), \hat{x}(t-12), \hat{x}(t-6), \hat{x}(t)) \quad (15)$$

and this is not what our initial model has been trained for.

Suppose a toy example of a series $x(t)_{t=1,4} = \{1, 2.5, 3.5, 4\}$. A model that tries to predict $\hat{x}(t+1) = f(x(t))$ would use as training set the pairs $[1; 2.5]$, $[2.5; 3.5]$, $[3.5; 4]$. Thus, using least-squares, a simple linear model that uses a single constant parameter $\hat{x}(t+1) = a * x(t)$ would solve the equation

$$[1, 2.5, 3.5]' * a = [2.5, 3.5, 4]'. \quad (16)$$

The solution is $a = 1.29$ and the MSE = 0.6. But in fact, the long term prediction training error, starting from the initial value $x(t_0) = 1$ and predicting the following values using the model output recursively, is MSE = 2.72 (the “long term” prediction would be $[1.29, 1.29^2, 1.29^3]$). Then, the right way to train a recursive long term prediction model would be

$$[1, 1 * a, 1 * a^2]' * a = [2.5, 3.5, 4]', \quad (17)$$

whose solution is $a = 1.67$ and the “long term” prediction MSE is 0.54. Nevertheless, even for this toy example, this is a nonlinear optimization problem with very high complexity, and thus the traditional function approximation modelling techniques do not work here.

5. Improving long term prediction using one-step-ahead prediction models

In the previous section, it was barely seen the overwhelming recursive task that would be to model directly a long term prediction problem from an I/O data set. As seen in [10], in principle the one-step-ahead prediction model can be directly used to perform that task. This section presents a general methodology to improve one-step-ahead predictors, when being used as recursive predictors.

5.1. Using the TaSe TSK fuzzy model

Consider the Mackey–Glass time series, and the optimal TaSe model obtained as shown in Table 1. Consider using this one-step-ahead predictor to perform long term predictions over the Mackey–Glass time series. Suppose then that an initial part of the series is known, and that the objective is to predict the rest using this model (at least a total of 23 (18 + 5) samples would be required, as an initial known piece of the series to predict the rest of the 500 values). Thus from this known initial piece, the estimated outputs would iteratively be used as new inputs (see Eq. (15)). Fig. 1 shows the long term prediction (Eq. (15), with $\hat{x}(t) = x(t)$ for the first values $t = 1, \dots, 23$) for the test data set. The RMSE obtained for the training data set equals 0.030, and for the test data set is 0.031.

These results show that a pretty good solution has been obtained for this long term prediction problem, using recursively a one-step-ahead predictor. Thanks to the accuracy of the TaSe optimal model, a sequence of outputs very similar to the original one was obtained.

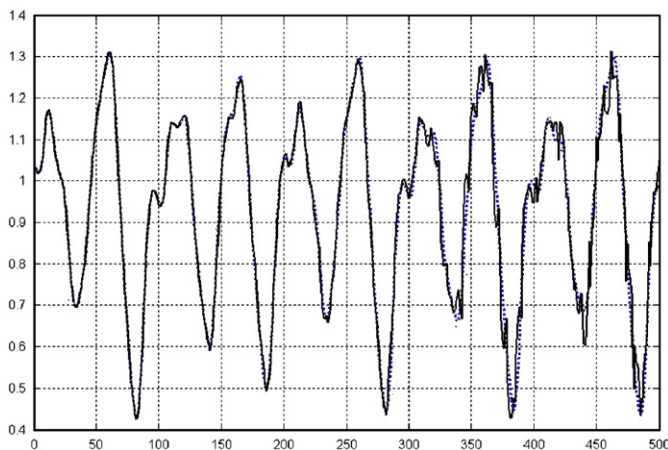


Fig. 1. Original (dotted line) and approximated (solid line) long term prediction for the test data set using the TaSe model. Test RMSE = 0.031.

Nevertheless, a modification in the training stage that will allow us to obtain better results is presented now. In [2], Schoelkopf et al. expanded the training data by adding small modifications in each training item for a specific written-digits classification problem. Similarly the time series training data for this case could be expanded, so that small modifications in time are also considered in the training. It could be possible, for example, to add a new set of samples obtained from the original piece of time series, by calculating $x(t + 0.5)$. But this is an interpolation problem and the task in some cases is at least as difficult as the prediction problem. Nevertheless, for the Mackey–Glass time series, a very good one-step-ahead approximation of the time series has been obtained (training RMSE = 0.0011). Thus, using the same optimal model, the training data were expanded to consider a new section of the time series, that is a small modification of the original time series which, in fact, shows the same behavior. The expanded set of data samples for training thus consist of the original $x(t), t = 1, \dots, 500$ plus a new fake piece of the series $\hat{x}, t = 1, \dots, 478$, where the $\hat{x}$ have been obtained using the TaSe predictive model as in Eq. (8).

For the retraining of the TaSe model using the expanded training set, note that only the consequent coefficients have been retrained. The structure and antecedent parameters remain the same, in order to avoid excessive computational cost and for the sake of simplicity. Fig. 2 shows the long term prediction for the test data set, obtained with the re-trained optimal TaSe model using the expanded training data set. The RMSE obtained for the long term prediction of the training data set is 0.015, and for the test data set equals 0.019. Further trials to re-extend the training data, by repeating the same process (re-extending the training data, using the one-step-ahead output of the retrained model), didn't bring further significant improvements for the long term prediction for this example.

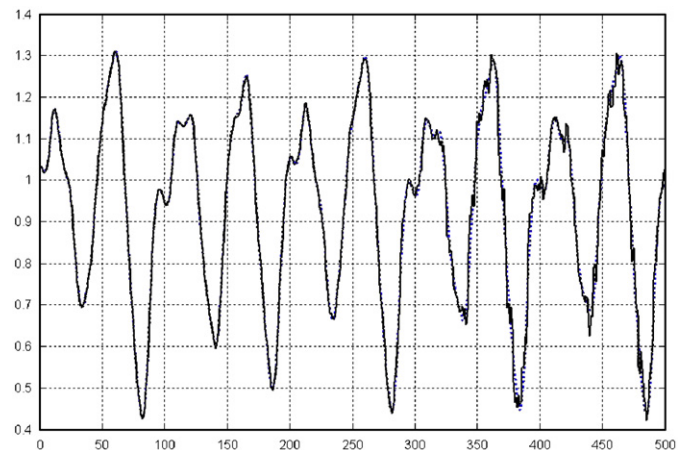


Fig. 2. Original (dotted line) and approximated (solid line) long term prediction for the test data set using the expanded training data set. Test RMSE = 0.019.

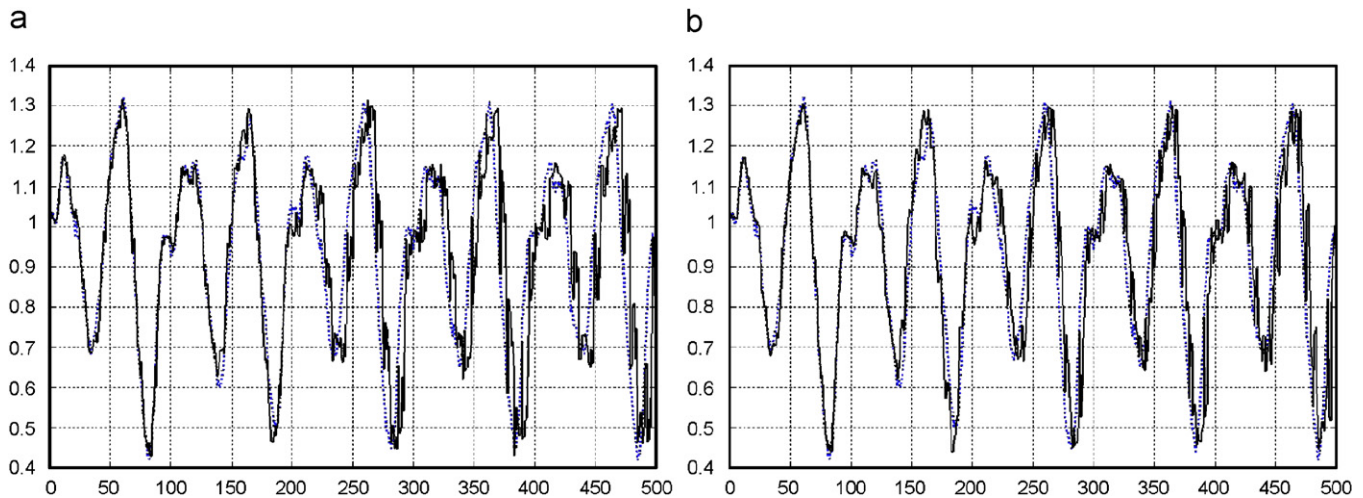


Fig. 3. (a) Original (dotted line) and approximated (solid line) long term prediction for the test data set using the original training data set for the noisy problem. Test RMSE = 0.13. (b) Original (dotted line) and approximated (solid line) long term prediction for the test data set using the expanded training data set for the noisy problem. Test RMSE = 0.10.

Now, consider the original Mackey–Glass time series and an added noise component with variance $0.05 * \sigma_x$, where σ_x is the standard deviation of the time series. Using this new example, the TaSe model was again used to learn a one-step-ahead predictor. The optimal TaSe model had in this case only 12 rules (the optimization was performed similarly as is explained in Section 2 for the noiseless Mackey–Glass series), due to the noisy behavior and to the use of a validation data set to avoid overfitting. As it was done for the noiseless series, the model was retrained using the output of the one-step-ahead prediction as new input data. But in this case, an iterative process was performed, repeating this technique. The output coming from Eq. (8) was used once again as new training data, obtaining new retrained one-step-ahead models. Fig. 3 shows the long term prediction for the test data set using the non-extended training data (Fig. 3a), with RMSE 0.12 and 0.13 for the training and test data sets. The iterative process was carried out until the long term prediction of the training data did not improve. The RMSE obtained after this iterative process was 0.10 for both data sets. The long term prediction of the retrained optimal TaSe model for the test data set is shown in Fig. 3b. This iterative process therefore shows to shortly improve the accuracy of the rule consequents for long term predictions in some cases.

The obtained results show that this process can be seen as a time series de-noising process. If a one-step-ahead model that is accurate enough is found, and it is able to partially get rid of the noise of the series, that model output can be used as training data, in order to obtain better results for recursive long term predictions.

As a second example, consider now a different time series, the monthly river flow series in cubic feet per second of the Snake river taken from [24]. Fig. 4 shows the shape of the series. Note how in this case, the series presents a high level of noise, although with a recognizable pattern in the series behavior.

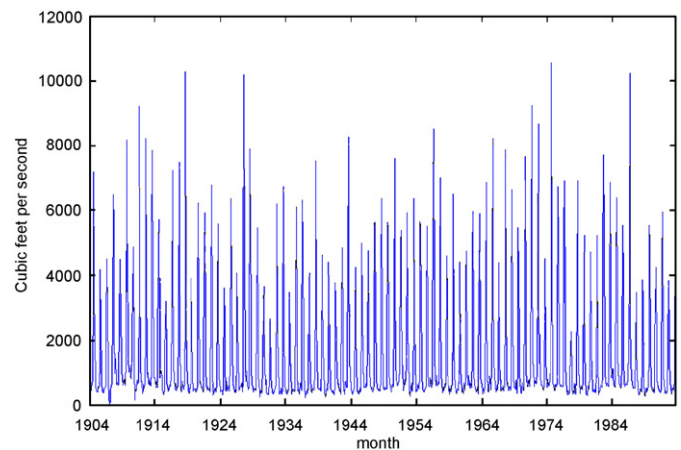


Fig. 4. Monthly river flow series in cubic feet per second of the Snake river, in the period of 1904 until 1994, see [24].

Again for this time series, 1000 samples were taken, 500 for training and validation (choosing randomly 400 and 100, respectively) and 500 for testing. In this case, the values taken for $h, \tau_0, \dots, \tau_n$ in the general equation $\hat{x}(t+h) = f(x(t-\tau_0), x(t-\tau_1), \dots, x(t-\tau_n))$ are [3, 0, 3, 6, 9] (see [16] on how to find an adequate value for these parameters on a given time series). The optimal TaSe model for this one-step-ahead prediction problem has only two rules, with training RMSE = 878 and test RMSE = 890. Note that since the same error measure has been used for this example (the RMSE), the errors in this non-normalized noisy series are much higher than for the previous example. The long term prediction for the training and test sets provide a total RMSE equal to 1769 for training and 1235 for test. Fig. 5 shows the long term prediction for the monthly Snake river flow, using the one-horizon-ahead TaSe predictor. It can be noted that the long term prediction ability of the one-horizon-ahead model vanishes within a short term.

Again for this time series example, the training data were extended with the output of the TaSe one-step-ahead prediction, considered as a new fake piece of the time series. This fake piece was added to the training data, keeping the previous optimal TaSe model structure and only reobtaining the optimal rule consequents. After one iteration, the long term prediction improves to training RMSE = 1618 and test RMSE 1060. Fig. 6 shows the long term prediction for the test and training data sets, using the extended training data set for the monthly Snake river flow. It the improvement that is obtained in the long term time series prediction is evident.

Again an iterative process was performed, extending the training data set, until the long term prediction of the original training data set did not improve. This way, the final long term training RMSE reached 1611 and the long term test RMSE 1041.

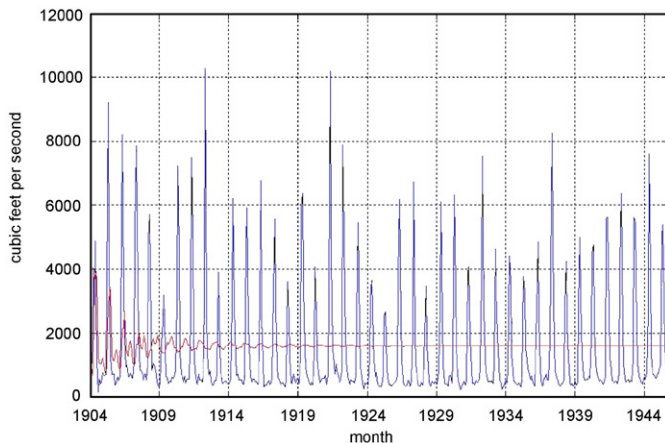


Fig. 5. Original (dotted line) and approximated (solid line) long term prediction for the training data set. Training RMSE = 1769.

5.2. Using LS-SVMs

Now consider the LS-SVM approach, and again the Mackey–Glass time series. Taking the series and using the one-step-ahead LS-SVM predictor to perform a recursive long term time series prediction lead to a training RMSE of 0.014 and test RMSE of 0.017. Again the same procedure was performed to extend the training data set with the output of the one-horizon-ahead LS-SVM predictor. The same calculated hyper-parameters $\sigma = 0.655$ and $\gamma = 4864$ were used for the retraining of the model with the extended data set. The long term prediction using the extended training data set improves up to RMSE = 0.0103 for the training data set, and RMSE = 0.0053 for the test data set.

Again with the Snake river flow time series, the same procedure was performed. The optimal one-horizon-ahead predictor obtained had as hyper-parameters $\sigma = 3.64$ and $\gamma = 7.62$. The training and test RMSE obtained were 910 and 898, respectively. The long term prediction was performed using recursively the one-horizon-ahead predictor, leading to a long term prediction of the training data set with an RMSE of 2292, and of the test data set with an RMSE of 1961. Then the training data set was extended with the one-step-ahead approximation of the training time series. The same hyper-parameters were used to retrain the LS-SVM one-step-ahead predictor (with the extended training data set). The long term prediction obtained for the training data set RMSE was 1598, and for the test data set RMSE was 1054. Fig. 7 shows the long term prediction for the test and training data sets, using the extended training data set for the monthly Snake river flow using the LS-SVM approach.

Then the iterative process was performed, generating more training data by obtaining new one-step-ahead predictions of the training data set, with the retrained LS-SVM model. For these further re-trainings, the

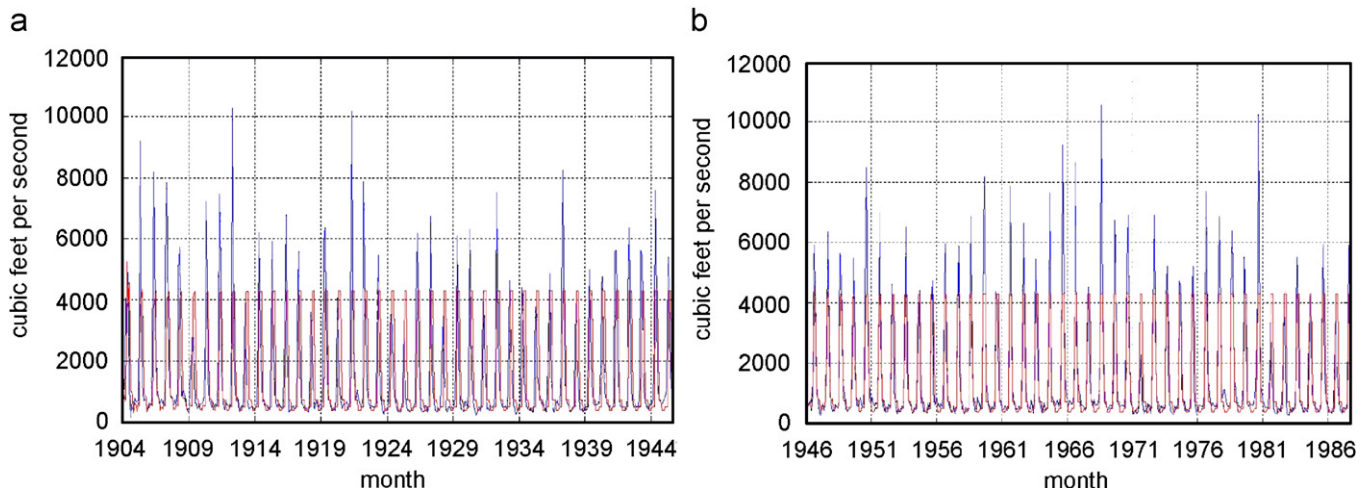


Fig. 6. (a) Original (dotted line) and approximated (solid line) long term prediction for the training data set using the extended training data set for river flow problem. Training RMSE = 1618. (b) Original (dotted line) and approximated (solid line) long term prediction for the test data set using the expanded training data set for the river flow problem. Test RMSE = 1060.

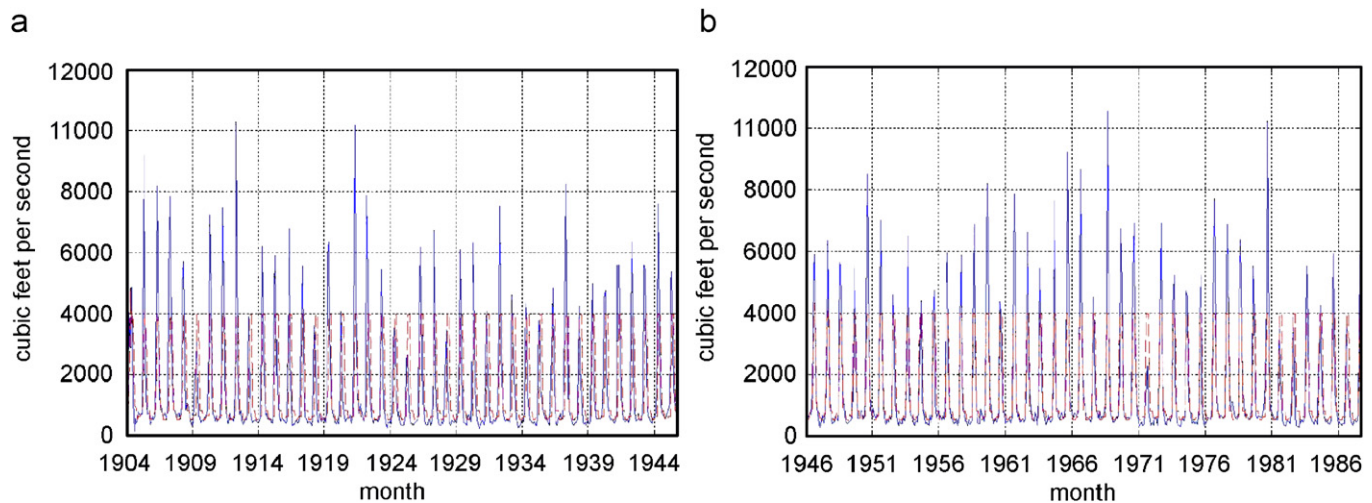


Fig. 7. (a) Original (dotted line) and approximated (solid line) long term prediction for the training data set using the extended training data set for river flow problem. Training RMSE = 1598. (b) Original (dotted line) and approximated (solid line) long term prediction for the test data set using the expanded training data set for the river flow problem. Test RMSE = 1054.

recursive long term prediction RMSE shortly reduced until 4 iterations more. Nevertheless, for LS-SVMs, this proposed process increases the size of the training data set, thus leading to a computational bottleneck in the training of the LS-SVM. The computational complexity of the LS-SVM training increases squarely with the number of data points in the training data [21]. Several approaches exist in the literature to overcome the computational bottleneck in kernel methods as the number of data points increases [20,21]. For example, fixed-size LS-SVM is an approach to deal with large-scale problems using LS-SVMs [21]. Other approaches can be found in [12].

6. Conclusions

This paper presented a general technique to improve the results on recursive prediction for long term time series prediction. To introduce and test this method, two completely different learning methodologies have been applied, the TaSe TSK fuzzy model and the LS-SVM model, to two different time series examples, the well-known artificial time series Mackey–Glass and a river flow time series. In general, when a one-step-ahead prediction of a given time series is performed, with any prediction model, a noiseless copy of the original series is obtained. The simulations have shown how the performance of the one-step-ahead predictor can be improved when it is used to perform recursive long term predictions. This improvement is obtained by using the noiseless one-step-ahead prediction as a new fake piece of the time series, which is added to the training data, and by retraining the model with this extended training data set. In the simulations, both models have improved their recursive long term prediction, for both time series. The LS-SVM approach obtains, in the simulations performed, excellent approximation results, thanks to its high level of flexibility and the control of the

bias-variance through the regularization parameter. However, the LS-SVM model suffers from the curse of dimensionality in the number of points and in principle lacks interpretability. The TaSe model obtains an accurate and interpretable model, comprising a set of rules whose consequents are the Taylor series expansion of the model output around the rule centers. However, as a model that uses grid input space partitioning, it can hardly be applied to complex problems with a high number of input variables. As further work, two main trends to obtain better recursive predictors will be followed. On the one hand, we intend to explore other possible variations to the general technique presented in Section 5. On the other hand, we are analyzing possible efficient optimizations for the general problem presented in Section 4.

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